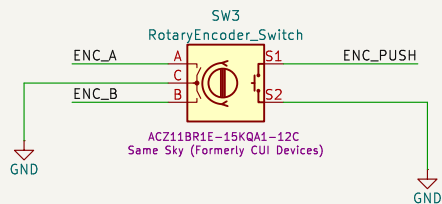


EC11 Encoder



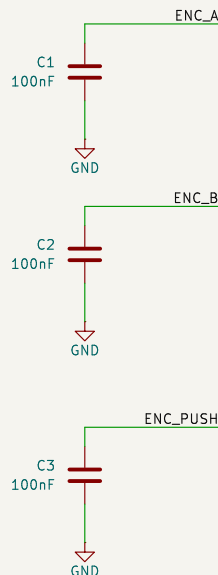
Debounce

Dev Note: This breakout debounce approach is relying on the internal pullups of the microcontroller.

In this case, I am assuming it's an nrf52, where according to a forum discussion the internal pullups is around 14kOhms.

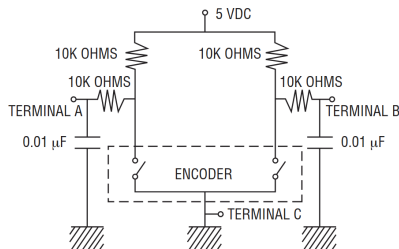
I chose 100nF because it's a cheap capacitance value.

devzone.nordicsemi.com/f/nordic-q-a/107277/nrf52832-pull-up-pull-down-resistance

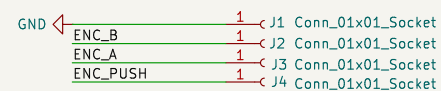


For proper filtering
<https://www.bourns.com/docs/Product-Datasheets/PEC11R.pdf>
 Bourn has a recommended filtering schematic shown here:

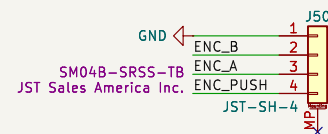
Suggested Filter Circuit



Socket



Dev Note: This allows for point to point wire soldering



Dev Note: This is using the same connector as a common i2c qwiic connector.

Intended for use with https://github.com/mofosyne/roTec_pcb

Sheet: /
 File: ec11_horizontal_breakout_board.kicad_sch

Title: EC11 Horizontal Rotary Encoder Breakout Board

Size: A4 Date: 2025-12-27

KiCad E.D.A. 9.0.6

Rev: 2

Id: 1/1